MILITARY SPECIFICATION

SHOT, HVAPDS-T; 76/50MM, M331A1; METAL PARTS

1. SCOPE

1.1 This specification covers hypervelocity armor-piercing discarding-sabot (HVAPDS) shot for use in 76MM Gun, T91 and T124.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND PUBLICATIONS

2.1 The following specifications, drawing and publication, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

QQ-M-151 - Metals; General Specification for Inspection of

MILITARY

MIL-G-2550 - General specification for Ammunition Except Small-arms Ammunition
MIL-P-10025 -Packing and Marking for Domestic Shipment of inert Ammunition Components; General Specification for

DRAWINGS

ORDNANCE CORPS

P-82082 Shot, HVAPDS-T, 76/50MM, M331A1, Loading Assembly and Marking Diagram.

ORDNANCE CORPS PUBLICATION

STANDARD INSPECTION PROCEDURE

ORD-SIP-M1-110 Shot, HVAPDS-T; 76/50MM, M331A1, Parts

(Copies of specifications, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 General.-The metal parts shall comply with all requirements specified on the drawing and applicable specifications and shall have no defects that would make the material hazardous for use.

3.3 Proving ground (production lot).

3.3.1 Security.

3.3.1.1 The complete shot (see 6.3) shall withstand firing in a service-able gun (serviceability of gun to be determined by the Proof Director), with a propellant charge adjusted to obtain 112 percent of the maximum rated pressure in a new gun, at target 1000 yards distance from the muzzle of the gun.

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3.3.1.2 The sheath and core assembly with T5E2 Tracer shall separate from the sabot (sleeve-carrier assembly) before reaching a paper screen placed approximately 100 feet from the muzzle of the gun.

3.3.1.3 There shall be no malfunctioning of the sabot in the gun tube, nor shall there be any sub-projectile metal component failure during flight to the target.

3.3.2 Ramming.-The sub-projectile shall not separate from the sabot when subjected to a ramming test.

3.3.3 Plate Penetration.-The sub-projectile shall completely penetrate (see 6.4) 3-inch homogenous (Class B) armor plate placed approximately 300 feet from the muzzle of the gun at 55° obliquity from the vertical.

3.3.4 Acceptance Provisions.-A lot of shot will be considered to have complied with these requirements if samples selected and tested in accordance with 4.7 comply with the acceptance provisions specified therein.

3.4 Pilot lot.

3.4.1 Size of lot and its disposition.-The contractor shall produce and submit a pilot-lot sample of at least 40 metal-parts sub-assemblies to the Contracting Officer or representative designated by him, for metallurgical, ballistic and other tests in order to determine the satisfactoriness of his methods and materials. The pilot lot shall be given 100-percent inspection by gaging and other non-destructive tests. Of this lot, 38 sub-assemblies shall be assembled by the contractor into complete shot, (see 6.5), and forwarded to the Proving Ground for ballistic tests (see 4.6). The remaining 2 sub-assemblies shall be retained at the inspection arsenal for destructive metallurgical examination as deemed necessary. Production continued under the contract, prior to the approval of the pilot lot, will be at the risk of the contractor.

3.4.2 Proving ground.

3.4.2.1 Plate penetration.-The sub-projectile shall completely penetrate (see 6.4) 3-inch homogenous (Class B) armor plate, placed approximately 300 feet from the muzzle of the gun at 55° obliquity from the vertical; and 7-inch homogenous (Class B) armor plate placed approximately 300 feet from the muzzle of the gun at 30° obliquity from the vertical.

3.4.2.2 Security.

3.4.2.2.1 The sheath and core assembly with T5E2 Tracer shall separate from the sabot (sleeve-carrier assembly) before reaching a paper screen placed approximately 100 feet from the muzzle of the gun.

3.4.2.2.2 There shall be no malfunctioning of the sabot in the gun tube, nor shall there be any sub-projectile metal-component failure during flight to the target.

3.4.2.3 Ramming.-The sub-projectile shall not separate from the sabot when subjected to a ramming test (see 4.6.1.3).

3.5 Workmanship.

3.5.1 Cleanliness.-All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

3.5.2 Soundness of cores.-Cores shall be free of warpage, surface pits, cracks, or other surface irregularities of such magnitude that would cause the cores to fail to conform to mechanical, physical, and dimensional requirements.

3.5.3 Processing.-Cores shall be manufactured by a commercially acceptable cold-pressing or hot-pressing process and shall comply with all requirements of dimensions and weight as specified on the drawing.

3.5.4 Coating and marking.-The sub-assemblies shall be coated in such a manner as to produce a tightly adherent coating of uniform thickness over the entire specified areas. The components shall be stenciled or marked as specified on the drawing. All painting and marking shall be dry to the touch before sub-assemblies are packed.

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4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Lot.

4.1.1 Definition. A lot shall consist of all assemblies manufactured by one manufacturer under one contract, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision. Each lot shall contain:

Cores made by one manufacturer at the same time, under the same conditions, which shall contain:

Tungsten-carbide powdered-metal mixture from not more than 10 batches

Binder of not more than one type (i.e. cobalt, or a nickel-cobalt mixture, etc.)

4.1.2 Miscellaneous lots.-Component parts left over from previously accepted lots may be processed and assembled into miscellaneous lots not exceeding 200. The quantity of shot submitted in miscellaneous lots shall not exceed five percent of the total quantity in the contract. A miscellaneous lot of shot shall be composed of cores that have been segregated into one of the three following general categories and so identified:

Cores containing no defects, but in which the powder batch cannot be identified, or cores containing no defects but that have otherwise been separated from their original lots.

Reprocessed cores originally containing either cracks or defects which appeared to be cracks.

Reprocessed cores that have dimensional or weight deviation or which originally contained surface defects that could not be classified as cracks. There will be no limitations as to the number of powder batches represented in the cores of a miscellaneous lot, but data cards shall describe the character of original defects in the cores and the approximate number of powder batches therein. Data cards shall also outline the procedure followed in reprocessing the cores where applicable.

4.1.3 Retention of identification marks.-A quantity of cores shall be assigned to a specific group of metal-parts sub-assemblies for identity as a shot lot. Cores shall be marked on the base by etching or equivalent means with core material batch identification and lot number of shot assembly. If, after being tested, the cores are rejected they shall be specially marked, reprocessed and reinspected. (See 4.4.3)

4.1.4 Use of metal-parts sub-assemblies.-Metal-parts sub-assemblies of shot lots that are rejected for faults attributable to the cores only may be reused in subsequent shot lots.

4.2 Sampling.-Number of samples, acceptance and rejection criteria used for determination of lot acceptance for tests specified in 4.5 shall be in accordance with Standard Inspection Procedure ORD-SIP-M1-110. Samples for proving ground test, paragraphs 4.6 and 4.7 shall be selected by the inspector in accordance with Table I.

TABLE I

Proving Ground Samples

	Pilot Lot	Production Lot
Security	10	5
Velocity	10	1[a]
Plate penetration (see 6.6)	18	9

[a] One sample shot from each production lot shall be retained by the manufacturer until five samples have been accumulated. These samples shall then be shipped to the proving ground for test.

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4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550 and Federal Specification QQ-M-151.

4.4 **Tests of core.**

4.4.1 Hardness test.-Each core shall be tested on the base or on the cylindrical portion approximately at the center, to determine compliance with the hardness requirement specified on the drawing for the core. A spot may be ground on the cylindrical portion not to exceed .010 inch deep. The hardness of cores shall be shown on the data card, listing the quantity of cores in each range of not more than two points Rockwell "A".

4.4.2 Bend test.-Each core shall be tested for gross flaws by subjecting to a bending load in the manner specified on the applicable drawing. In the case of minor superficial flaws, the core shall be positioned with the flaw in tension during the application of load.

4.4.3 Chemical tests.-A certified chemical analysis of the blended powder produced shall be furnished on the data card by the contractor (see 6.10)

4.4.4 Warpage.-Each core shall pass freely through a tube of the dimensions specified on the drawing.

4.5 Rotating-band hardness test.-Sample shot shall be selected and subjected to a hardness test using a Rockwell or similar-type hardness-testing machine to determine whether or not the bands have been softened to the extent required by the drawing.

4.6 Proving-ground test (pilot lot).

4.6.1 Procedure.

4.6.1.1 Plate penetration.-The sample shot, selected in accordance with Table I of 4.2, shall be fired from a gun for which the shot is standard, against 3-inch, (Class B) homogenous armor plate placed in a 55° obliquity from the vertical, and against 7-inch (Class B) homogenous armor plate placed in a 30° obliquity from the vertical. Place the armor plate approximately 300 feet from the muzzle of the gun. Determine the "calibration" value (see 6.7) of the armor plate by using shot of a reserved reference lot of the same caliber and model as the samples being tested. The Protection Ballistic Limit (see 6.8) of the plate using the pilot lot samples shall then be determined. In the event that more then six shot are fired for determination of Protection Ballistic Limit BL(P), the six impacts that shall be considered are those obtained with the three partial penetrations at the highest striking velocities and the three complete penetrations obtained at the lowest striking velocities. An impact that occurs under the following conditions shall be disregarded and another sample fired in its stead. (Cost of such additional samples and testing will be borne by the government.)

(1) Any core which strikes within three core diameters of the plate.
(2) Any core that strikes in the spalled area of a previous impact.
(3) Any core that strikes within two core diameters of a previous impact or whose path of penetration intersects a previous hole.

4.6.1.2 Security test.-The sample shot, selected in accordance with Table I of 4.2, shall be assembled into complete rounds using the standard propelling charge established at 70°F for the projectile-gun combination. Five rounds shall be conditioned at a temperature of 125°F, and five rounds shall be conditioned at a temperature of -40°F, for a period of not less than 24 hours. Within five minutes after removal from the conditioning chamber, the shot shall be fired through paper screens placed 100 and 300 feet from the gun, into a target at approximately 1000 yards, range. For comparison purposes shot from a reserved reference lot of the same caliber and model as the lot undergoing test shall be fired alternately with the sample shot. Observe for separation of the sub-projectile from the sabot before reaching the first paper screen, malfunctioning of the gun tube, and/or sub-projectile metal-component failures during flight to the target.

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4.6.1.3 Ramming test.-One sample round, selected from the samples to be used for velocity accuracy test (4.6.1.4) shall be rammed three times into the gun chamber for which it is standard, at the fastest speed obtainable by hand or automatic loading mechanism, if available. Note if separation of the sub-projectile from the sabot occurs.

4.6.1.4 Velocity test.-The sample shot, selected in accordance with Table I of 4.2, shall be fired from a gun for which standard for comparison with a group of ten shot from a reference lot, utilizing the service charge. Fire the sample shot and reference shot alternately, and obtain velocity and accuracy measurements for information only (see 6.9) at approximately 100 yards range. Record accuracy in terms of maximum spread in inches, both horizontal and vertical, and Probable Error (P.E.) in mils horizontal and vertical.

4.6.2 Acceptance.-The pilot shall be considered acceptable provided no malfunctioning of the sabot occurs in the gun tube, no sub-projectile component failure occurs during flight to the target, no separation of the sub-projectile from the sabot occurs in the ramming test, and provided the Protection Ballistic Limit of the armor plate, with the pilot lot, is not greater than 100 fps. of the calibration value.

4.7 Proving ground test (production lot).

4.7.1 Procedure.

4.7.1.1 Plate penetration.-The procedure for test determination of calibration value and Protection Ballistic Limit of the armor plate shall be conducted in the same manner as specified in 4.6.1.1, except that production lot samples shall be used and firing shall be against 3-inch, (Class B) homogenous armor plate in a 55° obliquity from the vertical.

4.7.1.2 Ramming test.-The test shall be conducted in the same manner as specified in 4.6.1.3 except that a production lot sample, selected from the security-accuracy test (4.7.1.3), shall be used. Note if separation of the sub-projectile from the sabot occurs.

4.7.1.3 Security.-The sample shot, selected in accordance with Table I of 4.2, shall be fired in a serviceable gun for which standard, with a propellent charge adjusted to obtain 112% of maximum rated pressure in a new gun. The shot shall be fired through paper screens placed 100 feet and 300 feet from the gun, into a target approximately 1000 yards range. Observe for separation of the sub-projectile from the sabot before reaching the first paper screen and malfunctioning of the sabot in the gun tube, and/or sub-projectile metal component failures during flight to the target. Record accuracy (see 6.9) in terms of maximum spread in inches, both horizontal and vertical, and Probable Error (P.E.) in mils horizontal and vertical.

4.7.1.4 Velocity.-The sample shot, selected in accordance with Table I of 4.2, shall be fired from a gun for which standard for comparison with a group of ten shot from a reference lot, utilizing the service charge established for a new gun. Fire the sample shot and reference shot alternately, and obtain velocity and accuracy measurements for information only (see 6.9) at approximately 1000 yards range. Record accuracy in terms of maximum spread in inches, both horizontal and vertical, and Probable Error (P.E.) in mils horizontal and vertical.

4.7.2 Acceptance.-The lot shall be considered acceptable provided the shot fired comply with the requirements of 3.3.1, 3.3.2 and 3.3.3. The lot shall be considered eligible for retest under the provisions of 4.7.3 provided no malfunctioning of the sabot occurs in the gun tube, no sub-projectile component failure occurs during flight to the target, no separation of the sub-projectile from the sabot occurs in the ramming test, and provided:

the Protection Ballistic Limit of the armor plate is greater than 100 fps. of the calibration value, and/or

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not more than one sub-projectile fails to separate from the sabot before reaching the first paper screen in the security accuracy test.
The lot shall be rejected if the limits for eligibility for retest are exceeded.

4.7.3 Retest.

4.7.3.1 Plate penetration.-A retest as provided for in 4.7.2 may be made if requested by the contractor using eighteen sample shot. The Protection Ballistic Limit shall be determined twice, independently. The lot shall be considered acceptable on retest provided the Protection Ballistic Limit of the armor plate is not greater than 100 fps of the calibration value.

4.7.3.2 Security.-A retest as provided for in 4.7.2 may be made if requested by the contractor using ten sample shot. The lot shall be considered acceptable on retest provided the shot fired comply with the requirements of 3.3.1.1, 3.3.1.2 and 3.3.1.3.

(Note: The retest shall be limited to the phase in which failures occur, i.e. plate penetration, or security.)

4.8 Reprocessing.-In the event of failure attributable to cores in the plate penetration retest, the manufacturer may reprocess the cores in the lot by re-sintering, repressing, or reinspection and submit the lot for retest. A letter "A" shall be added to the lot number of reprocessed lots. The sample size shall be eighteen and the test shall be conducted in accordance with 4.7.1.1.

5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking.-Packing, labeling and marking shall be as specified in Specification MIL-P-10025.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550. In addition the data cards shall contain on the reverse side the following information:

1. Name of core manufacturer.
2. Name of sabot and other metal-parts manufacturer.
3. Chemical composition of core.
4. Hardness of cores in range of two points, Rockwell "A".
5. Method of core manufacture (hot-pressed, cold-pressed or other).
6. Details of reprocessing, when applicable.
7. Category, in accordance with 4.1.2, when applicable.

6. NOTES

6.1 Intended use.-See 1.1 Scope.

6.2 Aluminum surfaces.-Information should be obtained from the contracting officer concerning commercial products found satisfactory for preparing aluminum surfaces for painting when such surfaces come in contact with ferrous parts.

6.3 Complete shot.-A complete shot is defined as an assembly of the Base and Sleeve Assembly, Sheath and Core Assembly, Diaphragm and Belleville Washer, with an empty or loaded T5E2 Tracer (Tracer to be furnished by the Government).

6.4 A complete penetration is considered to have occurred after an impact that results in:

a. A fragment of the plate that is dislodged from the rear with sufficient energy to perforate a 1-inch plywood board mounted parallel to and 1 foot behind the plate. The effect of a dislodged fragment is assessed by the Proof Officer in the event plywood is not used, or

b. Any part of the projectile that passes through the plate. No regard is given the amount or condition of the projectile.

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6.4 Waiting for ballistic results.-The manufacturer may wait until results of ballistic test are known before proceeding with assembling of the shot in a lot.

6.5 Extra samples.-To cover the possibility of unfair impacts in the armor-plate penetration test and the necessity of obtaining replacement samples and the consequent delay in obtaining ballistic test results, additional samples for the plate penetration test may be shipped to the proving ground with the original sample. Unused samples will be returned to the contractor, or the contractor will be reimbursed for the cost samples used to replace those which impacted unfairly.

6.6 "Calibration value" is defined as the Protection Ballistic Limit BL(P) of the plate obtained by using shot of a reserved reference lot of the same caliber and model as the samples being tested.

6.7 "Protection ballistic limit" of armor plate is defined as the mean of six velocities, three complete and three partial penetrations, the velocity spread to be not more than 150 feet per second.

6.8 Velocity and accuracy requirements.-Proving-ground tests for velocity and dispersion are not acceptance tests, but the results will be scrutinized carefully by the Ordnance Corps to determine quality of the ammunition produced. Evidence of excessive variation in any phase of the proving-ground tests will result in immediate investigation by the Ordnance Corps. Firing of shot from each of the test lots shall be interspersed with one of a reference lot, continuing until test is completed.

6.9 "Data-card information".-While the chemical composition given on the applicable drawing is advisory rather than mandatory, a complete analysis will be included on the data card for informational purposes only. The composition given on the drawing represents that with which past experience indicated best results are obtainable.

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